Further Strategies — problem 14.18

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There are n weights on a table with weights $m_1 > m_2 > \cdots > m_n$ and a two-pan scale. The weights are put on the pans one-by-one. To each weighing we assign a word from the alphabet $\{L, R\}$. The kth letter of the word is L or R if the left or right pan outweighs the other, respectively. Prove that any word from $\{L, R\}$ can be realized.